

Wireless Transmission – Communication satellites – Telephone System: Structure, Local loop, Trunks and Multiplexing and Switching. Data Link Layer: Design Issues – Error Detection and Correction.

1.1 WIRELESS TRANSMISSION

- People want to connect their mobile devices such as laptop, notebook, shirt, pocket, palmtop or wristwatch computers without using wires.
- Wireless transmission means sending information (data, voice or video) from one place to another without using physical wires or cables.

1.1.1 The Electromagnetic Spectrum

- ❖ When electrons move, they create electromagnetic waves that can propagate through space (even in a vacuum). These waves were predicted by the British physicist James Clerk Maxwell in 1865 and first observed by the German physicist Heinrich Hertz in 1887.
- ❖ The number of oscillations per second of a wave is called its frequency, f , and is measured in Hz (in honor of Heinrich Hertz). The distance between two consecutive maxima (or minima) is called the wavelength, which is universally designated by the Greek letter λ (lambda).
- ❖ When an antenna of the appropriate size is attached to an electrical circuit, the electromagnetic waves can be broadcast efficiently and received by a receiver some distance away. All wireless communication is based on this principle.
- ❖ In a vacuum, all electromagnetic waves travel at the same speed, no matter what their frequency. This speed, usually called the speed of light, c , is approximately 3×10^8 m/sec, or about 1 foot (30 cm) per nanosecond.

The fundamental relation between f , λ , and c (in a vacuum) is $\lambda f = c$

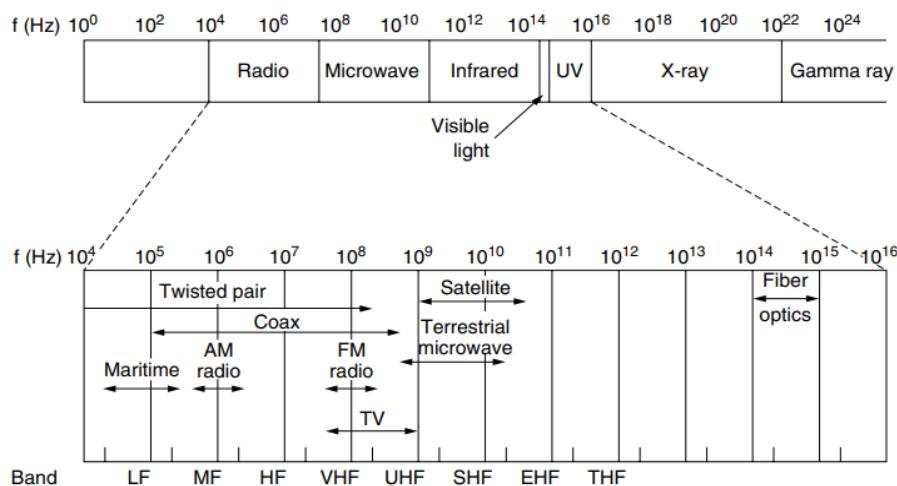


Figure 2-10. The electromagnetic spectrum and its uses for communication.

- ❖ Wireless transmission is a form of unguided media.
- ❖ Wireless communication involves no physical link established between two or more devices, connecting wirelessly.
- ❖ Wireless signals are spread over in the air and are received and interpreted by appropriate antennas.
- ❖ When an antenna is attached to electrical circuit of a computer or wireless devices, it converts the digital data into wireless signals and spread all over within its frequency range.

- ❖ The receptor on the other end receives these signals and converts them back to digital data.
- ❖ A little part of electromagnetic spectrum can be used for wireless transmission.

Frequency hopping spread spectrum:

- The transmitter hops from frequency to frequency hundreds of times per second.
- It is popular for military communication because it makes transmission hard to detect and next to impossible to jam.

Direct sequence spread spectrum:

- This spreads the signal over a wide frequency band.
- Some wireless LANs use this.

1.1.2 Radio Transmission

- ❖ Radio waves are easy to generate, can travel distances and can buildings easily, so they are widely used for communication both indoors and outdoors.
- ❖ Radio waves also are omnidirectional, meaning that they travel in all directions from the source, so the transmitter and receiver do not have to be carefully aligned physically.
- ❖ The properties of radio waves are frequency dependent. At low frequencies, radio waves pass through obstacles well.
- ❖ but the power falls off sharply with distance from the source—at least as fast as $1/r^2$ in air—as the signal energy is spread more thinly over a larger surface. This attenuation is called path loss.
- ❖ At high frequencies, radio waves tend to travel in straight lines and bounce off obstacles.
- ❖ They are also absorbed by rain.

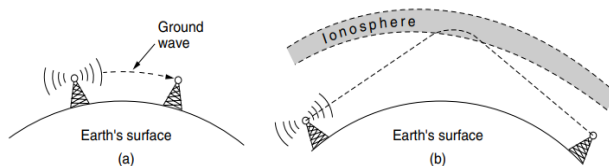


Figure 2-12. (a) In the VLF, LF, and MF bands, radio waves follow the curvature of the earth. (b) In the HF band, they bounce off the ionosphere.

1.1.2 Microwave Transmission

- ❖ Microwave communication is so widely used for long-distance telephone communication, mobile phones and televisions distribution.
- ❖ It has several significant advantages over fiber.
- ❖ The main one is that no right of way is needed. By buying a small plot of ground every 50 km and putting a microwave tower on it, one can bypass the telephone system and communicate directly.
- ❖ Microwave is also relatively inexpensive. Putting up two simple towers and putting antennas on each one may be cheaper than burying 50 km of fiber.

1.1.3 Infrared Transmission

- ❖ Unguided infrared waves are widely used for short-range communication. The remote controls used for televisions, VCRs, and stereos all use infrared communication.
- ❖ They are relatively directional, cheap, and easy to build but have a major drawback: they do not pass through solid objects. (Try standing between your remote control and your television and see if it still works.)

- ❖ Infrared communication has a limited use on the desktop, for example, to connect notebook computers and printers with the IrDA (Infrared Data Association) standard, but it is not a major player in the communication game.

1.1.4 Lightwave Transmission

- ❖ Wireless Lightwave transmission, known as Free-Space Optical(FSO) communication, extends this concept without physical mediums.
- ❖ Experiments demonstrate its feasibility in applications such as satellite communication, inter-building links, and even underwater data transfer.
- ❖ However, challenges such as atmospheric interfaces, line-of-sight requirements and precision alignment affect its reliability.
- ❖ A scenario illustrates the impact of environmental factors wireless Lightwave transmission.
- ❖ Data transmission works flawlessly at night but fails during the day due to sunlight interfaces and heat-induced air turbulence.
- ❖ The communication system must account for environmental variability and physical constraints. While Lightwave transmission is theoretically efficient, practical reliability and robustness under adverse conditions.

2.1 COMMUNICATION SATELLITES:

- ❖ Communication satellites have some interesting properties that make them attractive for many applications. In its simplest form, a communication satellite can be thought of as a big microwave repeater in the sky.
- ❖ It contains several transponders, each of which listens to some portion of the spectrum, amplifies the incoming signal, and then rebroadcasts it at another frequency to avoid interference with the incoming signal. This mode of operation is known as a bent pipe.
- ❖ The higher the satellite, the longer the period. Near the surface of the earth, the period is about 90 minutes.
- ❖ Consequently, low-orbit satellites pass out of view fairly quickly, so many of them are needed to provide continuous coverage and ground antennas must track them.
- ❖ At an altitude of about 35,800 km, the period is 24 hours. At an altitude of 384,000 km, the period is about one month.
- ❖ A satellite's period is important, but it is not the only issue in determining where to place it.
- ❖ Another issue is the presence of the Van Allen belts, layers of highly charged particles trapped by the earth's magnetic field.
- ❖ Any satellite flying within them would be destroyed fairly quickly by the particles. These factors lead to three regions in which satellites can be placed safely.

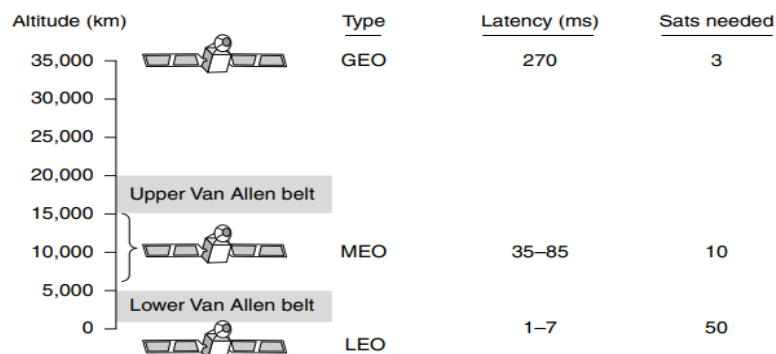


Figure 2-15. Communication satellites and some of their properties, including altitude above the earth, round-trip delay time, and number of satellites needed for global coverage.



2.1.1 Geostationary Satellites

- ❖ The first artificial communication satellite, Telstar, was launched in July 1962. Since then, communication satellites have become a multibillion dollar business.
- ❖ These high-flying satellites are often called GEO (Geostationary Earth Orbit) satellites.
- ❖ Modern satellites can be quite large, weighing over 5000 kg and consuming several kilowatts of electric power produced by the solar panels.
- ❖ The effects of solar, lunar, and planetary gravity tend to move them away from their assigned orbit slots and orientations, an effect countered by on-board rocket motors. This fine-tuning activity is called station keeping.
- ❖ A recent development in the communication satellite world is the development of low-cost microstations, sometimes called VSATs (Very Small Aperture Terminals).
- ❖ These tiny terminals have 1-meter or smaller antennas (versus 10 m for a standard GEO antenna) and can put out about 1 watt of power. The uplink is generally good for up to 1 Mbps, but the downlink is often up to several megabits/sec.
- ❖ In many VSAT systems, the microstations do not have enough power to communicate directly with one another (via the satellite, of course). Instead, a special ground station, the hub, with a large, high-gain antenna is needed to relay traffic between VSATs, as shown in Fig. 2-17.

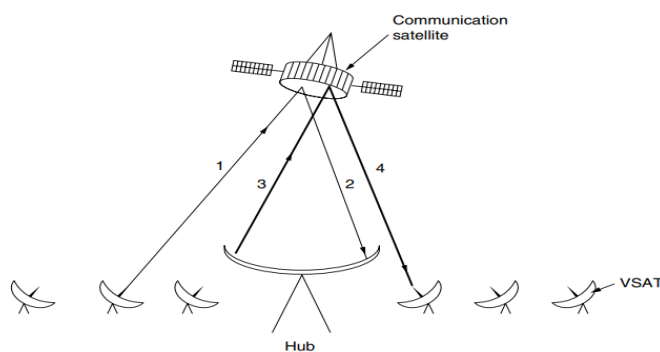


Figure 2-17. VSATs using a hub.

2.1.2 Medium-Earth Orbit Satellites

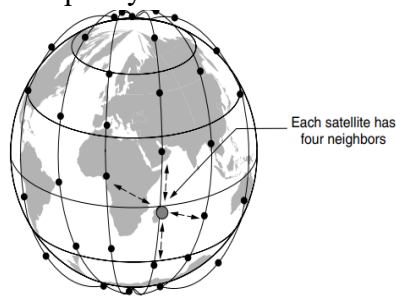
- ❖ At much lower altitudes, between the two Van Allen belts, we find the MEO (Medium-Earth Orbit) satellites.
- ❖ As viewed from the earth, these drift slowly in longitude, taking something like 6 hours to circle the earth.
- ❖ Accordingly, they must be tracked as they move through the sky.
- ❖ Because they are lower than the GEOs, they have a smaller footprint on the ground and require less powerful transmitters to reach them.
- ❖ Currently they are used for navigation systems rather than telecommunications, so we will not examine them further here. The constellation of roughly 30 GPS (Global Positioning System) satellites orbiting at about 20,200 km are examples of MEO satellites.

2.1.4 Low-Earth Orbit Satellites

- ❖ Moving down in altitude, we come to the LEO (Low-Earth Orbit) satellites. Due to their rapid motion, large numbers of them are needed for a complete system.
- ❖ On the other hand, because the satellites are so close to the earth, the ground stations do not need much power, and the round-trip delay is only a few milliseconds.
- ❖ In this section we will examine two examples of satellite constellations for voice service, Iridium and Globalstar.

Iridium

- ❖ The Iridium satellites are positioned at an altitude of 750 km, in circular polar orbits. They are arranged in north-south necklaces, with one satellite every 32 degrees of latitude, as shown in Fig. 2-18. Each satellite has a maximum of 48 cells (spot beams) and a capacity of 3840 channels.



Globalstar

- ❖ An alternative design to Iridium is Globalstar. It is based on 48 LEO satellites but uses a different switching scheme than that of Iridium.
- ❖ Whereas Iridium relays calls from satellite to satellite, which requires sophisticated switching equipment in the satellites, Globalstar uses a traditional bent-pipe design.

CubeSats

- ❖ They are satellites in units of 10 cm × 10 cm × 10 cm cubes, each weighing no more than 1 kilogram, that can be launched for as little as \$40,000 each. The launcher flies as a secondary payload on commercial space missions. It is basically a tube that takes up to three units of cubesats and uses springs to release them into orbit.

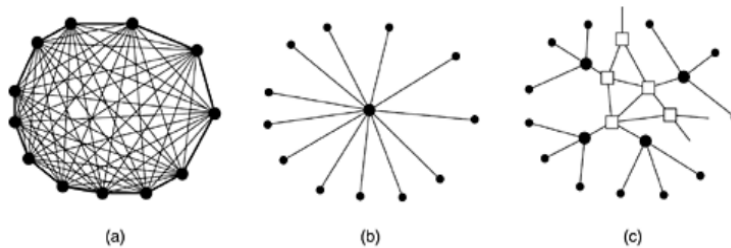
3.1.TELEPHONE SYSTEM

- ❖ PSTN (Public Switched Telephone Network), were usually designed many years ago, with a completely different goal in mind: transmitting the human voice in a more-or-less recognizable form.
- ❖ let us make a rough but illustrative comparison of the properties of a typical computer-computer connection via a local cable and via a dial-up telephone line.
- ❖ A cable running between two computers can transfer data at 109 bps, maybe more. In contrast, a dial-up line has a maximum data rate of 56 kbps, a difference of a factor of almost 20,000.

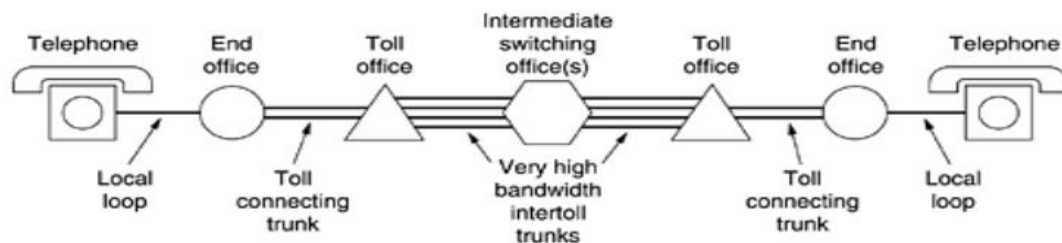
3.1.1 Structure of the Telephone System

- ❖ after Alexander Graham Bell patented the telephone in 1876 , there was an enormous demand for his new invention. The initial market was for the sale of telephones, which came in pairs.
- ❖ If a telephone owner wanted to talk to n other telephone owners, separate wires had to be strung to all n houses. as shown in Fig. 2-20(a)
- ❖ he cities were covered with wires passing over houses and trees in a wild jumble.
- ❖ To solve this problem, Bell saw this and formed the Bell Telephone Company, which opened its first switching office in 1878.
- ❖ The company ran a wire to each customer's house or office. To make a call, the customer would crank the phone to make a ringing sound in the telephone company office to

attract the attention of an operator, who would then manually connect the caller to the callee by using a jumper cable. The model of a single switching office is illustrated in Fig. 2-20(b).



- ❖ people wanted to make long distance calls between cities, so the Bell system began to connect the switching offices.
- ❖ To connect every switching office to every other switching office by means of a wire between them quickly became unmanageable, so second-level switching offices were invented.
- ❖ After a while, multiple second level offices were needed, as illustrated in Fig. 2-20(c).
- ❖ Eventually, the hierarchy grew to five levels., A telephone network consisting only of telephones (the small dots) end offices(the large dots) and toll offices(the squares).



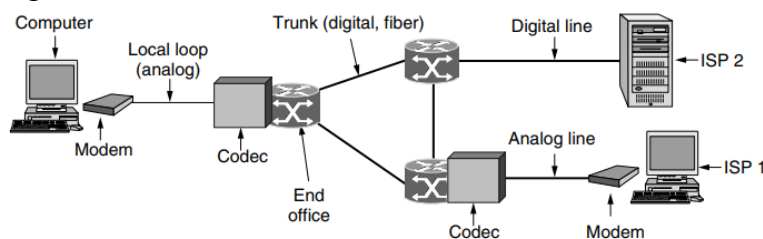
The telephone system consists of three major components:

1. Local loops (analog twisted pairs going into houses and businesses).
2. Trunks (digital fiber optics connecting the switching offices).
3. Switching offices (where calls are moved from one trunk to another).

.1.3 The Local Loop: Modems, ADSL, and Wireless

Telephone Modems

- ❖ A device that converts between a stream of digital bits and an analog signal that represents the bits is called a modem, which is short for “modulator demodulator.” Modems come in many varieties: telephone modems, DSL modems, cable modems, wireless modems, etc.
- ❖ The modem may be built into the computer or be a separate box . Logically, the modem is inserted between the (digital) computer and the (analog) telephone system, as seen in Fig. 2-32



- ❖ Telephone modems are used to send bits between two computers over a voice-grade telephone line, in place of the conversation that usually fills the line.
- ❖ A modem converts digital bits into analog signals for transmission and converts received analog signals back into digital bits.
- ❖ The term modem stands for modulator–demodulator.
- ❖ Telephone modems transmit data over voice-grade telephone lines instead of voice signals.
- ❖ Voice-grade telephone lines are limited to about 3100 Hz bandwidth, which restricts data rates.
- ❖ According to the Nyquist theorem, the maximum symbol rate is about 6000 baud, but practical modems use 2400 baud.
- ❖ Higher data rates are achieved by using multiple bits per symbol through larger constellations.
- ❖ Trellis Coded Modulation (TCM) is used to provide error correction in high-speed modems.
- ❖ The maximum speed of standard telephone modems is 33.6 kbps, limited by the Shannon capacity of the telephone line.

Digital Subscriber Lines (DSL / xDSL)

- ❖ DSL provides high-speed digital communication over existing telephone local loops.
- ❖ Unlike telephone modems, DSL removes the 3100 Hz bandwidth limitation by bypassing the voice filter.
- ❖ DSL makes use of the full capacity of the local loop, which supports frequencies up to about 1 MHz.
- ❖ The available spectrum is divided into 256 channels, each of 4312.5 Hz using DMT (Discrete Multitone) modulation.
- ❖ Channel 0 is reserved for POTS, and channels 1–5 are unused to prevent interference.
- ❖ DSL bandwidth decreases as the distance from the end office increases.
- ❖ ADSL (Asymmetric DSL) allocates more bandwidth for downstream than upstream transmission.
- ❖ The DSLAM at the end office collects data from multiple DSL lines and forwards packets to the ISP.

Fiber To The Home (FttH)

- ❖ FttH uses optical fiber instead of copper for the local loop to provide very high data rates.
- ❖ Fiber local loops are passive, requiring no powered equipment between the home and the end office.
- ❖ Multiple homes share a single fiber connection using optical splitters and combiners.
- ❖ This architecture is known as a Passive Optical Network (PON).
- ❖ Downstream and upstream data are transmitted using different wavelengths.
- ❖ PONs can support high data rates over distances of up to 20 km.
- ❖ GPON and EPON are two common types of PON standards.
- ❖ Upstream transmission uses time slot allocation controlled by the end office to avoid collisions.

4.1 TRUNKS AND MULTIPLEXING

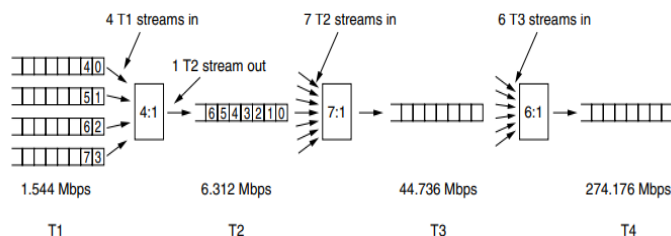
- ❖ Trunks in the telephone network are much faster than local loops.
- ❖ The core telephone network carries digital information (bits), not analog voice.
- ❖ Analog voice from local loops is converted to digital form at the end office.
- ❖ Trunks carry thousands or millions of calls simultaneously.
- ❖ Sharing trunks provides economies of scale in cost and maintenance.
- ❖ Trunk sharing is achieved using TDM and FDM multiplexing.
- ❖ High-bandwidth trunks cost nearly the same as low-bandwidth trunks to install.
- ❖ Multiplexing allows efficient use of trunk bandwidth.

Digitizing Voice Signals

- ❖ Early telephone networks carried voice as analog signals using FDM.
- ❖ FDM required analog circuitry and was not computer-friendly.
- ❖ Modern networks use TDM, which works entirely with digital electronics.
- ❖ Voice signals are digitized at the end office using a codec.
- ❖ The codec samples the signal 8000 times per second (125 μ sec/sample).
- ❖ Each sample is quantized into an 8-bit number.
- ❖ This digitization method is called PCM (Pulse Code Modulation).
- ❖ The standard uncompressed voice data rate is 64 kbps.

Time Division Multiplexing (TDM)

- ❖ TDM uses PCM samples to carry multiple voice calls over trunks.
- ❖ A voice sample from each channel is sent every 125 μ sec.
- ❖ Different countries use different incompatible TDM standards.
- ❖ T1 carrier is used in North America and Japan.
- ❖ T1 multiplexes 24 voice channels into a single stream.
- ❖ Each T1 frame contains 193 bits including a framing bit.
- ❖ The T1 data rate is 1.544 Mbps.
- ❖ TDM allows multiple carriers to be combined into higher-order carriers.



SONET / SDH

- ❖ SONET stands for Synchronous Optical Network.
- ❖ It was developed to allow interworking between different carriers.
- ❖ SONET provides a standard TDM system for optical fiber.
- ❖ The basic SONET frame has 810 bytes sent every 125 μ sec.
- ❖ The base data rate of SONET (STS-1) is 51.84 Mbps.
- ❖ SONET is a synchronous system controlled by a highly accurate clock.
- ❖ Frames are transmitted continuously, even if no data is present.
- ❖ SONET supports operations, administration, and maintenance (OAM).

SONET		SDH	Data rate (Mbps)		
Electrical	Optical	Optical	Gross	SPE	User
STS-1	OC-1		51.84	50.112	49.536
STS-3	OC-3	STM-1	155.52	150.336	148.608
STS-12	OC-12	STM-4	622.08	601.344	594.432
STS-48	OC-48	STM-16	2488.32	2405.376	2377.728
STS-192	OC-192	STM-64	9953.28	9621.504	9510.912
STS-768	OC-768	STM-256	39813.12	38486.016	38043.648

Figure 2-40. SONET and SDH multiplex rates.

Wavelength Division Multiplexing (WDM)

- ❖ WDM is a form of frequency division multiplexing for optical fiber.
- ❖ Multiple signals are transmitted using different wavelengths on the same fiber.
- ❖ Optical combiners merge signals of different wavelengths onto one fiber.
- ❖ At the receiver, optical splitters and filters separate the wavelengths.
- ❖ Each wavelength acts as an independent communication channel.
- ❖ systems are passive and highly reliable.
- ❖ Aggregate bandwidth increases with the number of wavelengths used.
- ❖ WDM enables extremely high data rates over long-haul fibers.

SWITCHING

- ❖ From the point of view of the average telephone engineer, the phone system is divided into two principal parts:
 - outside plant (the local loops and trunks, since they are physically outside the switching offices) .
 - inside plant (the switches, which are inside the switching offices).
- ❖ Two different switching techniques are used by the network nowadays:
 - circuit switching
 - packet switching.

Circuit Switching

- ❖ when you or your computer places a telephone call, the switching equipment within the telephone system seeks out a physical path all the way from your telephone to the receiver's telephone.
- ❖ This technique is called circuit switching. It is shown schematically in Fig. 2-42(a).

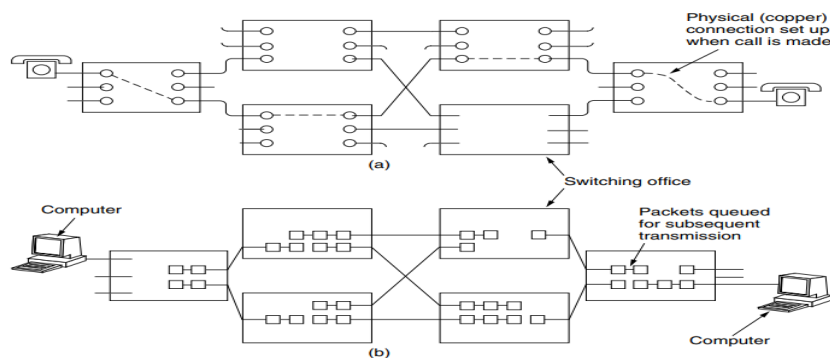


Figure 2-42. (a) Circuit switching. (b) Packet switching.

- ❖ Each of the six rectangles represents a carrier switching office (end office, toll office, etc.).
- ❖ In this example, each office has three incoming lines and three outgoing lines. When a call passes through a switching office, a physical connection is (conceptually)

established between the line on which the call came in and one of the output lines, as shown by the dotted lines.

Packet Switching

- ❖ The alternative to circuit switching is packet switching, shown in Fig. 2- 42(b)

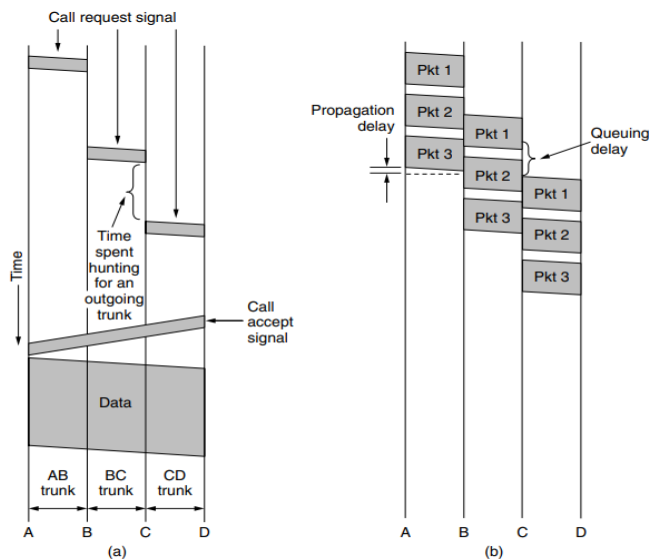


Figure 2-43. Timing of events in (a) circuit switching, (b) packet switching.

- ❖ It is up to routers to use store-and-forward transmission to send each packet on its way to the destination on its own.
- ❖ Packet-switching networks place a tight upper limit on the size, allowing packets to be buffered in router in router main memory instead of on disk.
- ❖ Advantages of packets switching over message switching is shown in fig a,b
- ❖ The first packet of multi packet message can be forwarded before the second one has fully arrived, reducing delay and improving throughput.
- ❖ For these reasons, computer networks are usually packet switched, occasionally circuit switched, but never message switched.

Message switching

- ❖ An alternative switching strategy is message switching .
- ❖ When this form of switching is used, no physical path is established in advance between sender and receiver .
- ❖ Instead, when the sender has a block of data to be sent, it is stored in the first switching office(i.e.router) and then forwarded later, one hop at a time.
- ❖ Each block is received in its entirety, inspected for errors and then retransmitted. A network using this techniques, is called a store-and-forward network.
- ❖ The first electro mechanical telecommunication systems used message switching , namely, for telegrams.

5.1 DATA LINK LAYER

- ❖ The data link layer uses the services of the physical layer to send and receive bits over communication channels.
- ❖ It has a number of functions, including:
 1. Providing a well-defined service interface to the network layer.
 2. Dealing with transmission errors.
 3. Regulating the flow of data so that slow receivers are not swamped by fast senders.

- ❖ To accomplish these goals, the data link layer takes the packets it gets from the network layer and encapsulates them into frames for transmission.
- ❖ Each frame contains a frame header, a payload field for holding the packet, and a frame trailer, as illustrated in Fig. 3-1.

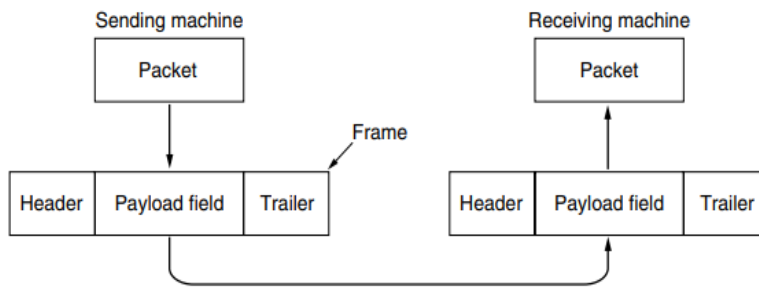


Figure 3-1. Relationship between packets and frames.

Services Provided to the Network Layer

- ❖ The function of the data link layer is to provide services to the network layer.
- ❖ The principal service is transferring data from the network layer on the source machine to the network layer on the destination machine
- ❖ The actual transmission follows the path of Fig. 3-2(b),

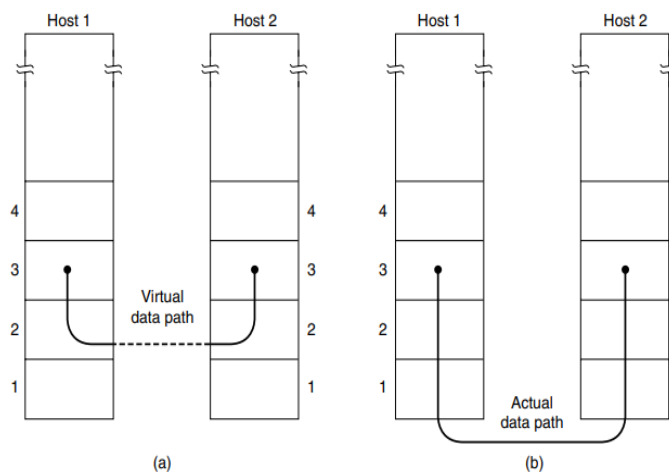


Figure 3-2. (a) Virtual communication. (b) Actual communication.

- ❖ The data link layer can be designed to offer various services. The actual services that are offered vary from protocol to protocol.
- ❖ Three reasonable possibilities that we will consider in turn are:
 1. Unacknowledged connectionless service.
 2. Acknowledged connectionless service.
 3. Acknowledged connection-oriented service.
- ❖ the source machine send independent frames to the destination machine without having the destination machine acknowledge.
- ❖ acknowledged connectionless service. When this service is offered, there are still no logical connections used, but each frame sent is individually acknowledged.
- ❖ In this way, the sender knows whether a frame has arrived correctly or been lost. If it has not arrived within a specified time interval, it can be sent again.
- ❖ When connection-oriented service is used, transfers go through three distinct phases.

- ❖ In the first phase, the connection is established by having both sides initialize variables and counters needed to keep track of which frames have been received and which ones have not.
- ❖ In the second phase, one or more frames are actually transmitted.
- ❖ In the third and final phase, the connection is released, freeing up the variables, buffers, and other resources used to maintain the connection.

Framing

- ❖ To provide service to the network layer, the data link layer must use the service provided to it by the physical layer.
- ❖ the data link layer to break up the bit stream into discrete frames, compute a short token called a checksum for each frame
- ❖ If the newly computed checksum is different from the one contained in the frame, the data link layer knows that an error has occurred and takes steps to deal with it
- ❖ The following four methods used to computer checksum error.

1. Character count.
2. Flag bytes with byte stuffing.
3. Flag bits with bit stuffing.
4. Physical layer coding violations.

- ❖ The first framing method uses a field in the header to specify the number of bytes in the frame.
- ❖ When the data link layer at the destination sees the byte count, it knows how many bytes follow and hence where the end of the frame is. This technique is shown in Fig. 3-3(a)

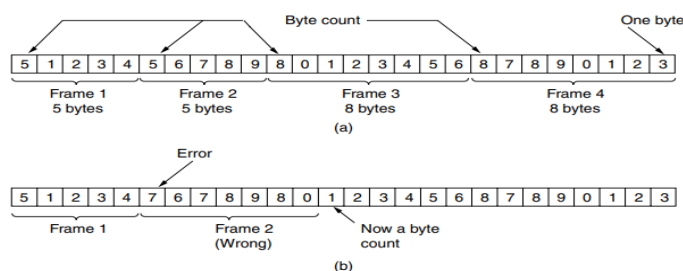


Figure 3-3. A byte stream. (a) Without errors. (b) With one error.

- ❖ The second framing method gets around the problem of resynchronization after an error by having each frame start and end with special bytes. Often the same byte, called a flag byte, is used as both the starting and ending delimiter.

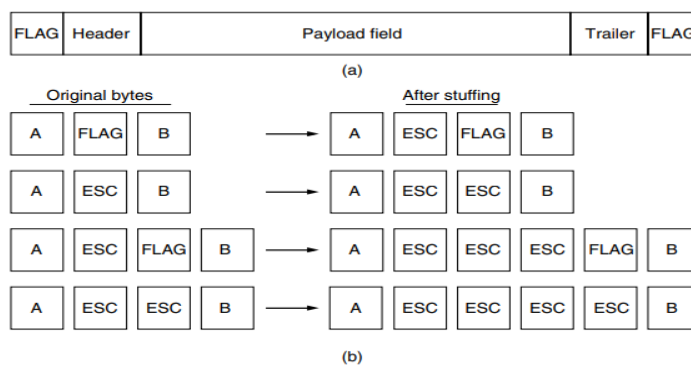


Figure 3-4. (a) A frame delimited by flag bytes. (b) Four examples of byte sequences before and after byte stuffing.

- ❖ The third method of delimiting the bit stream gets around a disadvantage of byte stuffing, which is that it is tied to the use of 8-bit bytes.
- ❖ Framing can also be done at the bit level, so frames can contain an arbitrary number of bits made up of units of any size. It was developed for the once very popular HDLC (Highlevel Data Link Control) protocol.

Error control

- ❖ Error control determines how to make sure all frames are eventually delivered to the network layer at the destination and in the proper order.
- ❖ If the sender receives a positive acknowledgement about a frame, it knows the frame has arrived safely. On the other hand, a negative acknowledgement means that something has gone wrong and the frame must be transmitted again.
- ❖ When the sender transmits a frame, it generally also starts a timer.
- ❖ If either the frame or the acknowledgement is lost, the timer will go off, alerting the sender to a potential problem. The obvious solution is to just transmit the frame again.
- ❖ When frames may be transmitted multiple times there is a danger that the receiver will accept the same frame two or more times and pass it to the network layer more than once.
- ❖ To prevent this from happening, it is generally necessary to assign sequence numbers to outgoing frames, so that the receiver can distinguish retransmissions from originals.

Flow control

- ❖ Another important design issue that occurs in the data link layer (and higher layers as well) is what to do with a sender that systematically wants to transmit frames faster than the receiver can accept them.
- ❖ This situation can occur when the sender is running on a fast, powerful computer and the receiver is running on a slow, low-end machine.
- ❖ Two approaches are commonly used.
 - In the first one, feedback-based flow control, the receiver sends back information to the sender giving it permission to send more data, or at least telling the sender how the receiver is doing.
 - In the second one, rate-based flow control, the protocol has a built-in mechanism that limits the rate at which senders may transmit data, without using feedback from the receiver.

ERROR DETECTION AND CORRECTION

- ❖ Error Is a condition when the output information does not match with the input information.
- ❖ During transmission, digital signals suffer from noise that can introduce errors in the binary bits travelling from one system to another.
- ❖ That means a 0 bit may change to 1 or a 1 bit may change to 0.

Error-Correcting Codes

We will examine four different error-correcting codes:

1. Hamming codes.
2. Binary convolutional codes.
3. Reed-Solomon codes.

4. Low-Density Parity Check codes.